

ASSIGNMENT1
MSC INFORMATION TECHNOLOGY
REVIEW PAPER ASSIGNMENT
ACTIVITY: LITERATURE REVIEW AND REVIEW PAPER WRITING

Objective:

The purpose of this activity is to develop students' ability to conduct a rigorous literature review, critically analyze existing research, identify research gaps, and propose directions for future research in Information Technology.

Student Instructions

Each student should:

1. **Select one research topic** from the approved list of MSc IT research topics.
2. Conduct a comprehensive literature search using reliable academic databases such as **IEEE Xplore, ScienceDirect, SpringerLink, ACM Digital Library, Scopus, Web of Science, and Google Scholar**.
3. Review **at least 30 relevant peer-reviewed research papers** related directly to the selected topic.
4. Give preference to **recent publications**, particularly studies published within the last **5-7 years**, while including older seminal studies where necessary.
5. Critically compare the methodologies, datasets, technologies, findings, strengths, and limitations of the reviewed studies.
6. Identify **research gaps, inconsistencies, limitations, and emerging trends**.
7. Develop a review paper following an accepted academic structure and referencing style.
8. Use appropriate reference-management software such as **Zotero, Mendeley, or EndNote**.
9. Ensure that all sources are properly cited and that plagiarism is avoided.
10. Submit the final review paper together with a **literature matrix** containing information about the reviewed papers.

RECOMMENDED REVIEW PAPER STRUCTURE

1. Title

The title should be specific and clearly indicate the technology, problem, and review scope.

Example:

Machine Learning-Based Intrusion Detection Systems: A Systematic Review of Approaches, Datasets, Challenges and Future Directions

2. Abstract

The abstract should briefly present:

- Background
- Research problem
- Objective of the review
- Review methodology
- Number of papers reviewed
- Major findings
- Research gaps
- Future research directions

Recommended length: 200–300 words.

3. Keywords

Provide approximately **5–7 keywords**.

Example:

Machine Learning; Intrusion Detection; Cybersecurity; Network Security; Deep Learning; Anomaly Detection.

4. Introduction

The introduction should establish the research context.

Students should discuss:

4.1 Background

Explain the selected technology/problem and its importance.

4.2 Problem Statement

Explain what problem exists in the current body of knowledge.

4.3 Motivation for the Review

Explain why reviewing existing research is necessary.

4.4 Review Objective

Clearly state what the paper intends to achieve.

4.5 Research Questions

Students should formulate **3–5 review questions**.

For example:

- **RQ1:** What machine learning techniques have been used for intrusion detection?
- **RQ2:** What datasets are commonly used?
- **RQ3:** Which algorithms demonstrate the best performance?
- **RQ4:** What limitations exist in current approaches?
- **RQ5:** What research opportunities remain unexplored?

5. Review Methodology

This is one of the **most important sections** of the review paper.

Students should explain exactly how the literature was identified and selected.

5.1 Review Approach

Specify whether the study uses:

- Systematic Literature Review (SLR)
- Scoping Review
- Systematic Mapping Study
- Structured Literature Review

For this class, a **Systematic Literature Review (SLR)** is recommended.

5.2 Search Databases

Students should identify the databases used.

For example:

- IEEE Xplore
- ACM Digital Library
- SpringerLink
- ScienceDirect
- Scopus
- Web of Science
- Google Scholar

5.3 Search Strategy

Students should provide their search strings.

Example:

("machine learning" OR "deep learning") AND ("intrusion detection" OR "network intrusion detection") AND ("cybersecurity" OR "network security")

5.4 Inclusion Criteria

For example:

- Peer-reviewed papers
- Published in English
- Directly related to the research topic
- Published between 2019 and 2026
- Contains sufficient methodology and results
- Full text available

5.5 Exclusion Criteria

For example:

- Duplicate publications
- Non-peer-reviewed articles
- Conference abstracts without full papers
- Papers unrelated to the research questions
- Papers without sufficient methodological information

5.6 Screening Process

Students should demonstrate:

Identification → Screening → Eligibility → Final Inclusion

A **PRISMA flow diagram** should be used where appropriate.

6. Literature Classification and Analysis

Students should organize the 30+ papers into meaningful categories.

For example, for **AI-Based Fraud Detection**:

Category	Possible Classification
Algorithms	Random Forest, SVM, ANN, CNN, LSTM
Data	Banking, mobile money, credit cards
Detection method	Supervised, unsupervised, hybrid
Evaluation	Accuracy, Precision, Recall, F1-score
Environment	Cloud, mobile, web
Application	Banking, FinTech, e-commerce

7. Literature Review and Synthesis

This should **not simply summarize papers one by one**.

Students should compare and synthesize studies.

Weak approach

Smith et al. (2022) used Random Forest.

John et al. (2023) used SVM.

Peter et al. (2024) used CNN.

Strong approach

Recent studies demonstrate increasing adoption of machine learning for fraud detection, with Random Forest and ensemble-based approaches frequently achieving strong performance on structured transaction datasets. However, deep learning approaches such as CNN and LSTM have demonstrated advantages when dealing with complex temporal or sequential transaction patterns. Despite these advances, differences in datasets, feature engineering, evaluation metrics, and experimental environments make direct comparison difficult.

This demonstrates **critical synthesis rather than description**.

8. Comparative Analysis of the 30 Papers

Students should develop a literature matrix.

Recommended Literature Matrix

Author/Year	Objective	Method	Dataset	Key Findings	Limitations
Author 1	Detection	Random Forest	CICIDS	High accuracy	Dataset limitation
Author 2	Detection	CNN	UNSW-NB15	Good detection	High computation
Author 3	Detection	LSTM	NSL-KDD	Good temporal detection	Limited real-world data
...	...	...	...	...	...

At least **30 papers** should be included.

9. Thematic Analysis

Students should identify major themes emerging from the literature.

For example:

Theme 1: Artificial Intelligence Approaches

- Machine Learning
- Deep Learning
- Reinforcement Learning

Theme 2: Data Challenges

- Data imbalance
- Data quality
- Lack of labelled datasets

Theme 3: Model Performance

- Accuracy
- Precision
- Recall
- F1-score

Theme 4: Deployment Challenges

- Computational requirements
- Explainability
- Privacy
- Real-time processing

10. Research Gaps

This section is **essential for an MSc-level review paper**.

Students should identify gaps such as:

- Lack of real-world datasets
- Limited research in developing countries
- Limited Rwanda-specific studies
- Lack of explainable AI
- Poor interoperability
- Limited real-time implementation
- Limited comparative evaluation
- High computational requirements
- Privacy and ethical concerns
- Lack of standardized evaluation frameworks

The student should support every identified gap using evidence from the reviewed literature.

11. Proposed Research Framework

Based on the identified gap, students should propose a conceptual framework or research model.

For example:

Input Data → Data Preprocessing → Feature Engineering → AI Model → Threat/Fraud Detection → Explainability → Alert → Decision Support

The framework should be **derived from the literature**, not simply invented without theoretical justification.

12. Future Research Directions

Students should identify areas requiring further investigation.

For example:

- Integration of AI and blockchain
- Explainable AI
- Federated learning
- Real-time analytics
- Edge AI
- Privacy-preserving machine learning
- Rwanda-specific datasets
- Cross-sector cybersecurity frameworks

13. Conclusion

The conclusion should summarize:

- What the literature demonstrates
- Major findings
- Main limitations
- Major research gaps
- Future research opportunities

The conclusion should **not introduce completely new information**.

14. References

Students should provide complete references using a consistent referencing style.

Recommended:

IEEE or APA 7th Edition

The references should correspond exactly to the citations used in the paper.

Minimum Requirements

Requirement	Minimum
Peer-reviewed papers	30
Recommended recent papers	20+
Research questions	3-5
Academic databases	3+

Review period	2019–2026 recommended
Literature matrix	Required
PRISMA diagram	Recommended/required for SLR
Research gaps	Required
Future directions	Required
Referencing style	APA 7 or IEEE

Suggested Assessment Rubric – 100 Marks

Component	Marks
Topic definition and introduction	10
Review objectives and research questions	10
Literature search methodology	15
Quality and relevance of 30+ papers	15
Critical analysis and synthesis	20
Comparative literature analysis	10
Identification of research gaps	10
Future research directions	5
Referencing and academic writing	5
Total	100